ALMOST EVERYWHERE MONOTONE ADDITIVE FUNCTIONS

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Abstract. We prove that a real additive function which decreases on a set of natural density zero is a nonnegative constant multiple of the logarithm. This answers a conjecture of Erdős from 1946, listed as Problem 1122 on the Erdős Problems website. The main step is to deduce finite concentration without a growth assumption on the values at prime powers. A truncated normalization and a bounded clipping reduce this step to Mangerel's first-moment theorem for short intervals and the moment estimates of Ruzsa and Hildebrand.

1. Introduction

A function $f\colon \mathbb{N} \to \mathbb{R}$ is *additive* if $f(ab) = f(a) + f(b)$ whenever $(a, b) = 1$. Erdős proved that an additive function which is nondecreasing on $\mathbb{N}$ must be a constant multiple of $\log n$. He asked whether the same conclusion follows when the set of decreases has natural density zero [1, p. 3]. The question is also recorded as Erdős Problem 1122 [4].

Theorem 1.1. *Let $f\colon \mathbb{N} \to \mathbb{R}$ be additive. If*

$$D_f(X) := \#\{n \le X : f(n+1) < f(n)\} = o(X),$$

then there is a constant $c \ge 0$ such that $f(n) = c \log n$ for every $n \in \mathbb{N}$.

Mangerel established an approximate logarithmic representation under the same density hypothesis [3, Theorem 1.9]. For completely additive functions, he obtained the exact conclusion with additional assumptions on the values at primes and the number of decreases [3, Corollary 1.7]. His short-interval theorem is the analytic input used here.

We say that f has *finite concentration* if there are constants $d > 0$ and $R < \infty$ such that, for arbitrarily large X, some interval of length R contains at least dX of the values $f(n)$, $n \le X$. The interval is allowed to depend on X. Two results of Erdős reduce Theorem 1.1 to finite concentration. Theorem V of [1] states that finite concentration is equivalent to the existence of $c \in \mathbb{R}$ for which

$$\sum_p \frac{\min\{(f(p) - c\log p)^2, 1\}}{p} < \infty. \tag{1.1}$$

Theorem X of the same paper implies that, under (1.1), the limiting distribution of $f(n+1)-f(n)$ has no mass on $(-\infty, 0)$ only when $f(n) = c\log n$ for all n. Both results apply to general additive functions, with no restriction on their values at higher prime powers.

The proof below establishes finite concentration by contradiction. If it fails, we normalize f and subtract a logarithm so that the truncated quadratic mass of the prime values is a small fixed number M. The minimizing logarithmic coefficient supplies an orthogonality identity. We then compare a bounded clipping of the normalized function with a strongly additive function obtained by clipping its prime values. Their mean-square difference is small enough to retain a positive variance. On the other hand, the density hypothesis makes the bounded function almost constant on a typical interval of any fixed length. Short-interval averaging forces the variance of the strongly additive function to be smaller than its moment lower bound permits.

Date: September 7, 2026.

Notation. Letters p, q denote primes. We write

$$\mathbb{E}_X F = \frac{1}{X}\sum_{n\le X} F(n), \qquad \mathcal{A}_H F(n) = \frac{1}{H}\sum_{j=0}^{H-1} F(n-j) \quad (n \ge H).$$

Restricted averages, such as $\mathbb{E}_X(F\mathbf{1}_{n\ge H})$, retain the normalizing factor $1/X$. For a strongly additive function $z(n) = \sum_{p\mid n} a_p$, put

$$\mu_z(X) = \sum_{p\le X}\frac{a_p}{p}.$$

All implicit constants are absolute unless a dependence is indicated. The constants K and M used in the proof are fixed before X tends to infinity. In statements about a family z_X, the prime coefficients may depend on X.

2. Moment estimates and short intervals

For an additive function z, define

$$A_0(z,X) = \sum_{p^k\le X}\frac{z(p^k)}{p^k}, \qquad B_j(z,X)^j = \sum_{p^k\le X}\frac{|z(p^k)|^j}{p^k} \quad (j=2,4).$$

We use the following consequences of Hildebrand's moment theorem [2, Théorème 1*, equation (6)]:

$$\mathbb{E}_X|z - A_0(z,X)|^2 \asymp \inf_{b\in\mathbb{R}}\left(b^2 + \sum_{p^k\le X}\frac{|z(p^k) - b\log(p^k)|^2}{p^k}\right), \tag{2.1}$$

$$\mathbb{E}_X|z - A_0(z,X)|^4 \ll B_2(z,X)^4 + B_4(z,X)^4. \tag{2.2}$$

The second-moment equivalence (2.1) is due to Ruzsa; Hildebrand's theorem includes it as the case of exponent 2.

If z is strongly additive, then

$$|A_0(z,X) - \mu_z(X)| \le \sum_{p\le X}\frac{|a_p|}{p(p-1)} \ll \left(\sum_{p\le X}\frac{a_p^2}{p}\right)^{1/2}. \tag{2.3}$$

Also $B_2(z,X)^2 \le 2\sum_{p\le X} a_p^2/p$, with the analogous bound for B_4. Thus, if $|a_p| \le L$ and $\sum_{p\le X} a_p^2/p \le V$, then

$$\mathbb{E}_X|z - \mu_z(X)|^2 \ll V, \qquad \mathbb{E}_X|z - \mu_z(X)|^4 \ll V^2 + L^2V. \tag{2.4}$$

If the coefficients are bounded and $a_p \to 0$ for each fixed prime, dominated convergence in (2.3) gives $A_0(z,X) - \mu_z(X) \to 0$.

Lemma 2.1. *Let z_X be strongly additive functions satisfying*

$$|z_X(p)| \le L, \qquad \sum_{p\le X}\frac{z_X(p)^2}{p} \le V$$

for fixed $L, V < \infty$. Then

$$\lim_{H\to\infty}\limsup_{X\to\infty}\frac{1}{X}\sum_{H\le n\le X}|\mathcal{A}_H z_X(n) - \mu_{z_X}(X)|^2 = 0, \tag{2.5}$$

where H runs through the positive integers.

Proof. For $Y \le X$, write

$$d_Y = \frac{2}{Y} \sum_{Y/2<n\le Y} z_X(n), \qquad F_Y(n) = \mathcal{A}_H z_X(n) - d_Y.$$

Mangerel's Theorem 1.1 [3] gives, for every integer $10 \le H \le Y/100$,

$$\frac{2}{Y} \sum_{Y/2<n\le Y} |F_Y(n)| \ll \left(\sqrt{\frac{\log\log H}{\log H}} + (\log Y)^{-1/800}\right) \sqrt{2V}. \tag{2.6}$$

The estimate is uniform over all additive functions; in particular, it applies to the changing family z_X.

Counting multiples of each prime gives

$$d_Y - \mu_{z_X}(Y) = \sum_{p\le Y} z_X(p) \left(\frac{2}{Y}\left(\left\lfloor \frac{Y}{p}\right\rfloor - \left\lfloor \frac{Y}{2p}\right\rfloor\right) - \frac{1}{p}\right) = O\left(\frac{L\pi(Y)}{Y}\right). \tag{2.7}$$

Jensen's inequality and (2.4) at Y imply

$$\frac{2}{Y} \sum_{Y/2<n\le Y} |F_Y(n)|^4 \ll V^2 + L^2 V + O_L((\log Y)^{-4}). \tag{2.8}$$

To see this, center the backward average at $\mu_{z_X}(Y)$. Every index in its inner sums lies in $[1, Y]$ and occurs at most H times. The remaining constant is estimated by (2.7). Hölder's inequality now gives

$$\frac{2}{Y}\sum |F_Y|^2 \le \left(\frac{2}{Y}\sum |F_Y|\right)^{2/3} \left(\frac{2}{Y}\sum |F_Y|^4\right)^{1/3}.$$

Together with (2.6) and (2.8), this proves the required iterated limit on each fixed dyadic band $Y \asymp X$.

For completeness, this dyadic conclusion extends to the whole interval as follows. On each of finitely many bands with $Y \asymp X$, we have

$$|\mu_{z_X}(X) - \mu_{z_X}(Y)| \le L \sum_{Y<p\le X} \frac{1}{p} = o(1).$$

The omitted initial segment $H \le n \le \eta X$ contributes $O_{L,V}(\sqrt{\eta})$ to (2.5): apply Cauchy–Schwarz and Jensen, using the fourth-moment bound (2.4) at X. First take $X \to \infty$, then $H \to \infty$, and finally $\eta \downarrow 0$. ◻

3. A weighted estimate for a set of primes

Lemma 3.1. *Suppose* $0 < M < 1/4$ *and* $\mathcal{T}$ *is a set of primes at most* X *such that* $\sum_{p\in\mathcal{T}} 1/p \le M$. *Then, uniformly in* $\mathcal{T}$,

$$\sum_{p\in\mathcal{T}} \frac{1}{p} \sum_{X/p<q\le X} \frac{1}{q} \ll M\log(2/M) + o_{X\to\infty}(1). \tag{3.1}$$

Here M *is fixed in the term* $o(1)$.

Proof. For $p \le X^{1-M}$, the inner sum is at most $\log(1/M) + o(1)$ by the prime harmonic-sum estimate. These primes contribute at most $M\log(1/M) + o(1)$.

For $p > X^{1-M}$, discard the restriction $p \in \mathcal{T}$ and interchange the sums. The terms with $q > X^M$ are bounded by

$$\left(\sum_{X^{1-M}<p\le X} \frac{1}{p}\right)\left(\sum_{X^M<q\le X} \frac{1}{q}\right) \ll M\log(2/M) + o(1).$$

For $q \le X^M$, the condition $X/p < q$ implies $p > X/q$. The bound $\pi(t) \ll t/\log t$ and partial summation give, uniformly for $2 \le q \le X^M$,

$$\sum_{X/q<p\le X} \frac{1}{p} \ll \frac{\log q}{\log X}.$$

Their total contribution is therefore

$$\ll \frac{1}{\log X} \sum_{q \le X^M} \frac{\log q}{q} \ll M.$$

This proves (3.1). ∎

4. Normalization

In the remainder of the proof, suppose that f does not have finite concentration. For $s > 0$ define

$$W_X(s) = \min_{c\in\mathbb{R}} \left\{ c^2 + \sum_{p\le X} \frac{\min\{(f(p)/s - c\log p)^2, 1\}}{p} \right\}. \tag{4.1}$$

Lemma 4.1. *For each fixed $M > 0$ and all sufficiently large X, there are $s_X > 0$ and a minimizer c_X in (4.1) such that $W_X(s_X) = M$. They may be chosen so that $s_X \to \infty$; any such choices satisfy $c_X \to 0$. If*

$$u_p = \frac{f(p)}{s_X} - c_X \log p \quad (p \le X),$$

then

$$\sum_{p\le X} \frac{\min\{u_p^2, 1\}}{p} = M - c_X^2 = M - o(1), \tag{4.2}$$

$$\sum_{\substack{p\le X \\ |u_p|<1}} \frac{u_p \log p}{p} = c_X. \tag{4.3}$$

No u_p has absolute value exactly 1.

Proof. The expression in braces in (4.1) is continuous and coercive as a function of c, so its minimum is attained. The minimum is continuous in $s > 0$, since minimizers remain bounded when s varies over a compact subinterval. On writing $\lambda = sc$, the expression becomes

$$\frac{\lambda^2}{s^2} + \sum_{p\le X} \frac{\min\{(f(p) - \lambda\log p)^2/s^2, 1\}}{p}.$$

It follows that $W_X(s)$ is nonincreasing in s, and it tends to zero as $s \to \infty$ for fixed X.

For each fixed $s > 0$, we claim that $W_X(s) \to \infty$. Otherwise some unbounded sequence of X would have bounded $W_X(s)$ and bounded minimizers. Pass to a subsequence along which the minimizers converge to c. Fatou's lemma gives

$$\sum_{p} \frac{\min\{(f(p)/s - c\log p)^2, 1\}}{p} < \infty.$$

Rescaling by the fixed number s preserves convergence of the truncated square sum. Erdős's Theorem V would therefore give finite concentration of f, contrary to the assumption.

Continuity now supplies s_X with $W_X(s_X) = M$. Monotonicity and the preceding claim imply $s_X \to \infty$. We have $|c_X| \le \sqrt{M}$. If a subsequential limit of c_X were a nonzero number c, Fatou's lemma on the fixed primes would give

$$\sum_p \frac{\min\{(c\log p)^2, 1\}}{p} \le M,$$

which is impossible. Thus $c_X \to 0$, and (4.2) follows.

It remains to justify differentiation at the minimizer. At a point where $|f(p)/s_X - c\log p| = 1$, the derivative of the corresponding summand has a downward jump of size $2\log p/p$ as c increases. Coincident jumps add. A local minimum of a continuous piecewise differentiable function has left derivative at most zero and right derivative at least zero, so a downward jump excludes a local minimum. Consequently no such point is a minimizer. Differentiating (4.1) at c_X gives (4.3). □

5. Clipping the normalized function

Fix $K > 1$ and $0 < M < 1/4$, and use Lemma 4.1. Let $\phi_K(t) = \max\{-K, \min\{t, K\}\}$ and set

$$\mathcal{T} = \{p \le X : |u_p| > 1\}, \qquad S_0(n) = \sum_{\substack{p|n \\ p\notin\mathcal{T}}} u_p,$$

$$a_X = \sum_{\substack{p\le X \\ p\notin\mathcal{T}}} \frac{u_p}{p}, \qquad S(n) = S_0(n) - a_X.$$

Coefficients at primes greater than X are set to zero. Define

$$z_X(n) = \sum_{p|n} \phi_K(u_p), \qquad Z(n) = z_X(n) - a_X,$$
$$Y(n) = \phi_K\left(\frac{f(n)}{s_X} - c_X\log n - a_X\right). \tag{5.1}$$

The distinction is that Y clips the value of the additive function, whereas z_X clips each prime value before adding.

Lemma 5.1. *There is an absolute constant C such that*

$$\limsup_{X\to\infty} \mathbb{E}_X|Y - Z|^2 \le C\left(\frac{M}{K^2} + M^2\log(2/M) + K^2M^2\right). \tag{5.2}$$

Proof. The small coefficients have quadratic mass at most M, and $\sum_{p\in\mathcal{T}} 1/p \le M$. Hence

$$\mathbb{E}_X S^4 \ll M + M^2. \tag{5.3}$$

Put $T(n) = \#\{p \in \mathcal{T} : p \mid n\}$. Counting multiples of distinct primes gives

$$\mathbb{E}_X T(T-1) = \sum_{\substack{p,q\in\mathcal{T} \\ p\ne q}} \frac{1}{X}\left\lfloor\frac{X}{pq}\right\rfloor \le M^2. \tag{5.4}$$

We also need

$$\mathbb{E}_X(S^2\mathbf{1}_{T\ge 1}) \ll M^2\log(2/M) + o(1). \tag{5.5}$$

For $p \in \mathcal{T}$, the coefficient of p in S_0 is zero, so $S_0(pm) = S_0(m)$ for every m, including multiples of p. Writing $y = X/p$ and using (2.4) at y, we get

$$\begin{aligned}\mathbb{E}_X(S^2\mathbf{1}_{p|n}) &= \frac{1}{p}\frac{1}{y}\sum_{m\le y}|S_0(m) - a_X|^2 \\ &\ll \frac{1}{p}\left(M + |a_X - \mu_{S_0}(y)|^2\right).\end{aligned}$$

At $y = 1$, we have $S_0(1) = \mu_{S_0}(1) = 0$, so the same bound follows from the mean-shift term. By Cauchy–Schwarz,

$$|a_X - \mu_{S_0}(X/p)|^2 \le M \sum_{X/p<q\le X} \frac{1}{q}.$$

Now use $\mathbf{1}_{T\ge 1} \le \sum_{p\in\mathcal{T}} \mathbf{1}_{p|n}$, sum over $p \in \mathcal{T}$, and apply Lemma 3.1. This proves (5.5).

First replace the input to the clipping by the sum of its prime contributions, and write

$$Y_*(n) = \phi_K\left(S(n) + \sum_{\substack{p|n\\ p\in\mathcal{T}}} u_p\right).$$

If $T(n) = 0$, the difference $Y_*(n) - Z(n)$ vanishes for $|S(n)| \le K$ and otherwise has absolute value at most $|S(n)|$. Its squared contribution is at most $\mathbb{E}_X S^4/K^2$. If $T(n) = 1$, with unique tail value v, the Lipschitz property of ϕ_K gives

$$|\phi_K(S + v) - S - \phi_K(v)| \le 2|S|.$$

If $T(n) \ge 2$, boundedness of ϕ_K gives

$$|Y_*(n) - Z(n)|^2 \le CS(n)^2 + CK^2T(n)(T(n) - 1).$$

Equations (5.3)–(5.5) therefore imply

$$\mathbb{E}_X|Y_* - Z|^2 \ll \frac{M}{K^2} + M^2\log(2/M) + K^2M^2 + o(1). \tag{5.6}$$

We next compare Y and Y_*. If $p^k \| n$ with $k \ge 2$, its contribution to the difference between the two unclipped expressions is

$$\frac{f(p^k) - f(p)}{s_X} - c_X(k-1)\log p.$$

This tends to zero for each fixed prime power. Moreover, $\sum_{p,k\ge 2} p^{-k} < \infty$. Given $\varepsilon > 0$, choose a finite set of prime powers so that the sum of p^{-k} outside the set is less than ε. Uniformly in X, the proportion of integers divisible by any of those excluded prime powers is at most ε. On the complement, only the fixed finite set can contribute, and its total contribution tends uniformly to zero. Thus the unclipped difference tends to zero in probability. The function ϕ_K is bounded and Lipschitz, so

$$\mathbb{E}_X|Y - Y_*|^2 = o(1).$$

Together with (5.6), this proves the lemma. □

Lemma 5.2. *There are absolute constants $c_0, C_0 > 0$ such that*

$$\liminf_{X\to\infty} \mathbb{E}_X|z_X - \mu_{z_X}(X)|^2 \ge c_0M - C_0K^2M^2. \tag{5.7}$$

Proof. Write $v_p = \phi_K(u_p)$. Since $K > 1$, (4.2) implies

$$M - o(1) \le \sum_{p \le X} \frac{v_p^2}{p} \le K^2 M. \tag{5.8}$$

On $p \notin \mathcal{T}$ we have $v_p = u_p$, so (4.3) gives

$$\left| \sum_{p \le X} \frac{v_p \log p}{p} \right| \le |c_X| + KM \log X. \tag{5.9}$$

Using $\sum_{p \le X} (\log p)^2/p \sim \frac{1}{2}(\log X)^2$, we obtain

$$\begin{aligned} \inf_{b \in \mathbb{R}} \sum_{p \le X} \frac{(v_p - b \log p)^2}{p} &= \sum_{p \le X} \frac{v_p^2}{p} - \frac{\left(\sum_{p \le X} v_p \log p/p\right)^2}{\sum_{p \le X} (\log p)^2/p} \\ &\ge M - CK^2M^2 - o(1). \end{aligned} \tag{5.10}$$

Apply (2.1), dropping the nonnegative b^2 term and the terms with $k \ge 2$ from its right-hand side. For each fixed prime p, $v_p \to 0$, because $s_X \to \infty$ and $c_X \to 0$. Therefore (2.3) and dominated convergence give $A_0(z_X, X) - \mu_{z_X}(X) \to 0$. The second moments are uniformly bounded by (2.4) and (5.8); changing the center consequently changes the second moment by $o(1)$. Combining this with (5.10) proves (5.7). ☐

6. Proof of Theorem 1.1

Suppose first, as in Sections 4 and 5, that f has no finite concentration. The function Y in (5.1) is bounded between $-K$ and K. If $f(n+1) \ge f(n)$, monotonicity and Lipschitz continuity of ϕ_K imply

$$(Y(n) - Y(n+1))_+ \le |c_X| \log(1 + 1/n),$$

where $t_+ = \max\{t, 0\}$. At a decrease of f, the same quantity is at most $2K$. Telescoping the signed increments gives

$$\sum_{n < X} |Y(n+1) - Y(n)| \le 2K + 4KD_f(X) + 2|c_X| \log X = o(X). \tag{6.1}$$

For each fixed H, every difference $Y(n) - Y(n-j)$ is bounded by the sum of the j intervening absolute increments. Since $|Y| \le K$, (6.1) yields

$$\frac{1}{X} \sum_{H \le n \le X} |Y(n) - \mathcal{A}_H Y(n)|^2 \longrightarrow 0. \tag{6.2}$$

Let $m_X = \mu_{z_X}(X) - a_X$. Then $Z - m_X = z_X - \mu_{z_X}(X)$, and

$$Z - m_X = (Z - Y) + (Y - \mathcal{A}_H Y) + \mathcal{A}_H(Y - Z) + (\mathcal{A}_H Z - m_X). \tag{6.3}$$

The square of the right-hand side is at most four times the sum of the four squared terms. Jensen's inequality gives the contraction

$$\frac{1}{X} \sum_{H \le n \le X} |\mathcal{A}_H(Y - Z)(n)|^2 \le \frac{1}{X} \sum_{n \le X} |Y(n) - Z(n)|^2.$$

The prime coefficients of z_X are bounded by K and have quadratic mass at most K^2M, so Lemma 2.1 applies to the last term of (6.3). The first H indices contribute $o(1)$ to the centered second moment: this follows directly from (2.4) and Cauchy–Schwarz. Taking $X \to \infty$ and then $H \to \infty$ in (6.3), and using (6.2), we obtain

$$\begin{aligned} \limsup_X \mathbb{E}_X |z_X - \mu_{z_X}(X)|^2 &\leq 8 \limsup_X \mathbb{E}_X |Y - Z|^2 \\ &\leq C_1 \left(\frac{M}{K^2} + M^2 \log(2/M) + K^2 M^2 \right) \end{aligned} \tag{6.4}$$

for an absolute constant C_1.

Choose $K > 1$ so large that $C_1/K^2 < c_0/4$, where c_0 is from Lemma 5.2. Next choose $0 < M < 1/4$ so small that

$$C_1 M \log(2/M) + (C_1 + C_0) K^2 M < c_0/4.$$

With these fixed choices, (6.4) contradicts (5.7). Hence f has finite concentration.

Erdős's Theorem V now gives a constant c for which the additive remainder $r(n) = f(n) - c \log n$ satisfies

$$\sum_p \frac{\min\{r(p)^2, 1\}}{p} < \infty.$$

By his Theorem X, the differences $f(n+1) - f(n)$ have a limiting distribution, and this distribution has no negative mass only if r is identically zero. For each $\varepsilon > 0$, our hypothesis gives

$$\frac{1}{X} \#\{n \leq X : f(n+1) - f(n) < -\varepsilon\} \leq \frac{D_f(X)}{X} \longrightarrow 0.$$

The limiting distribution therefore has no mass on $(-\infty, 0)$. It follows that $f(n) = c \log n$ for all n. Finally, $c < 0$ would give a decrease at every integer, so $c \geq 0$. $\square$

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